

第九章 (牛顿/傅里叶)

$$R = \frac{1}{2\pi\lambda} \ln\left(\frac{d_2}{d_1}\right) \quad \text{单位长度热阻}$$

总热阻 = 单位长度热阻 × 管道长度 L_s

$$\text{集总参数法} \quad \tau_c = \frac{\rho c V}{h A} \quad \frac{\theta}{\theta_0} = e^{-\frac{h A}{\rho c V} \tau}$$

($Bi < 0.1$) ($Bi > 0.1$, 解析法)

$$\frac{\partial t}{\partial \tau} = a \frac{\partial^2 t}{\partial x^2} \quad a = \frac{\lambda}{\rho c}$$

1. 无量纲参数

- 毕渥数: $Bi = \frac{h\delta}{\lambda}$ (特征长度取平壁半厚 δ)
- 傅里叶数: $Fo = \frac{a\tau}{\delta^2} = \frac{\lambda\tau}{\rho c \delta^2}$
- 无量纲坐标: $X = \frac{x}{\delta}$ (x 为平壁内某点到中心的距离)

集总参数法

无限大平壁 厚 2δ	$Bi_v = \frac{h\delta}{\lambda}$	$Fo_v = \frac{a\tau}{\delta^2}$
无限长圆柱 半径 R	$Bi_v = \frac{h(R/2)}{\lambda}$	$Fo_v = \frac{a\tau}{(R/2)^2}$
圆球 半径 R	$Bi_v = \frac{h(R/3)}{\lambda}$	$Fo_v = \frac{a\tau}{(R/3)^2}$

注: 过余温度 ($\theta = t - t_0$)

$$\frac{\theta}{\theta_0} = \frac{t - t_f}{t_0 - t_f} = e^{-\frac{h A}{\rho V c} \tau}$$

$$\frac{t - 20}{250 - 20} = e^{-(0.001251 \times 300)}$$

上述方程解得:

$$Nu = \frac{hl}{\lambda}$$

$$Re = \frac{\rho u l}{\eta} = \frac{\rho u l}{\mu}$$

$$Pr = \frac{\nu}{a}$$

$$Eu = \frac{\Delta p}{\rho u^2}$$

$$Pe = Re Pr = \frac{ul}{a}$$

$$Gr = \frac{ag\Delta l^3}{\nu^2}$$

- 根据角系数的完整性:

$$F_{ab,cd} = 1 - F_{ab,ac} - F_{ab,bd}$$

$$F_{ab,ac} = \frac{ab + ac - bc}{2ab}$$

$$F_{ab,bd} = \frac{ab + bd - ad}{2ab}$$

由于垂直纸面方向的长度相同, 则有:

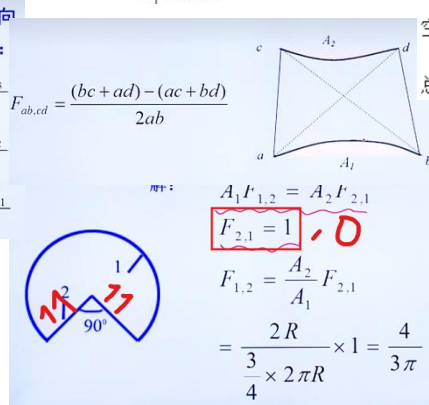
$$F_{1,2} = \frac{l_1 + l_2 - l_3}{2l_1}$$

$$F_{1,3} = \frac{l_1 + l_3 - l_2}{2l_1}$$

$$F_{2,3} = \frac{l_2 + l_3 - l_1}{2l_2}$$

由上式可解得

$$\epsilon'_1 = 0.854$$



$F_{2,2}$ 是不可能发射到自己, 单位宽度 $w=1$

$$\Phi_{1,2}^{(n)} = \frac{1}{n+1} \Phi_{1,2}$$

两个漫灰表面构成封闭空腔, $T_1 > T_2$, 表面1净损失、表面2净获得的热量分别为

$$\Phi_1 = \frac{E_{b1} - J_1}{\frac{1 - \epsilon_1}{A_1 \epsilon_1}} \quad \Phi_2 = \frac{J_2 - E_{b2}}{\frac{1 - \epsilon_2}{A_2 \epsilon_2}}$$

表面1、2之间净辐射换热量为

$$\Phi_{1,2} = A_1 X_{1,2} J_1 - A_2 X_{2,1} J_2 = A_1 X_{1,2} (J_1 - J_2) = \frac{J_1 - J_2}{\frac{1}{A_1 X_{1,2}}}$$

根据能量守恒,

$$\Phi_1 = \Phi_2 = \Phi_{1,2}$$

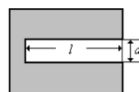
辐射热功率=内能减少速度

$$E_b = \sigma_b T^4 = 5.67 \times 10^{-8} \times 1400^4 =$$

由维恩位移定律得到:

$$\lambda_{\max} = \frac{\lambda_{\max} T}{T} = \frac{2897.6}{1400} = 2.07 \mu m$$

解: (1) 假设空腔开口为一温度为0K的黑体表面2, 则空腔内表面1与该黑体表面2间的辐射换热量为从空腔发射出的辐射能,



$$\Phi = \frac{E_{b1} - E_{b2}}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 X_{1,2}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}} = \frac{E_{b1} - 0}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 X_{1,2}} + 0}$$

$$A_2 = \frac{\pi d^2}{4} = \frac{\pi \times (0.1)^2}{4} = 0.00785 m^2$$

$$A_1 = \pi d l + \frac{\pi d^2}{4} = 0.1335 m^2$$

$$\Phi = \frac{\sigma T_1^4}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 X_{1,2}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}} = \frac{5.67 \times 10^4}{\frac{1 - 0.2}{0.1335 \times 0.2} + \frac{1}{0.00785 \times 1}} = 360.3 W$$

(2) 单位时间内空腔的净辐射散热损失

$$\Phi' = \frac{E_{b1} - E_{b2}}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 X_{1,2}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 X_{1,2}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}}$$

$$= \frac{5.67 \times (10^4 - 3^4)}{157.4} = 357.3 W$$

(3) 依题意, 设空腔内表面的发射率至少为 ϵ' , 则从空腔口发射的辐射能为

$$\frac{E_{b1}}{\frac{1 - \epsilon'_1}{A_1 \epsilon'_1} + \frac{1}{A_1 X_{1,2}}} = A_2 \epsilon_2 E_{b1}$$

$$A_1 X_{1,2} = A_2 X_{2,1} \quad \epsilon_2 = 0.99 \quad X_{2,1} = 1$$

由上式可解得

$$\epsilon'_1 = 0.854$$

$$Re = \frac{u_m d}{\nu} \leq 2300 \quad \text{层流}$$

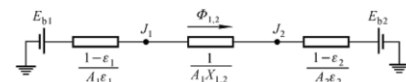
$$2300 < Re < 10^4 \quad \text{层流到湍流的过渡}$$

$$Re > 10^4 \quad \text{旺盛湍流}$$

联立以上三式, 可得

$$\Phi_{1,2} = \frac{E_{b1} - E_{b2}}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 X_{1,2}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}}$$

两表面封闭空腔的辐射网络



(基尔霍夫定律, 节点电流/热流=0)

由三个漫灰表面组成的封闭空腔的辐射换热网络

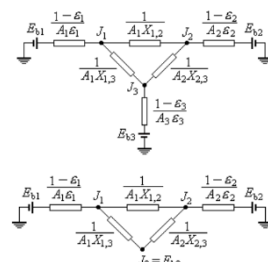
重辐射表面:

有效辐射等于投入辐射, $J_i = G_i$, 净辐射换热等于零。

对漫灰体重辐射面, $\epsilon_i = \alpha_i$, $J_i = \epsilon_i E_{bi} + (1 - \alpha_i) G_i$

重辐射面的存在改变了辐射能的方向分布, 所以影响整个系统的辐射换热。

黑体 $\epsilon=1$ 理想白体 $\epsilon=0$ 单位面积辐射换热 (q)



辐射网络中的重辐射面3

热量类型	表面热阻 $R_{\text{表}}$	空间热阻 $R_{\text{空}}$
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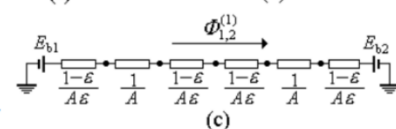
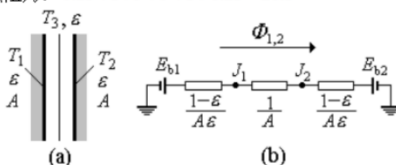
$$\text{单位面积热流密度 } q \quad \frac{1 - \epsilon}{\epsilon} \quad \frac{1}{X_{1,2}}$$

$$\text{总热流量 } \Phi = qA \quad \frac{1 - \epsilon}{A \epsilon} \quad \frac{1}{A X_{1,2}}$$

$$\frac{1 - \epsilon_1}{\epsilon_1} = \frac{1 - 0.8}{0.8} = 0.25, \quad \frac{1 - \epsilon_2}{\epsilon_2} = \frac{1 - 0.1}{0.1} = 9, \quad \frac{1 - \epsilon_3}{\epsilon_3} = \frac{1 - 0.8}{0.8} = 0.25$$

$$\text{空间热阻为 } \frac{1}{\Phi_{1,2}} = \frac{1}{\Phi_{1,2}}$$

$$\text{总热阻为: } 0.25 + 1 + 9 + 9 + 1 + 0.25 = 20.5$$

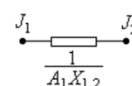
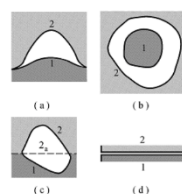


$$R_{\text{总}} = R_{\text{表1}} + R_{\text{空}} + R_{\text{表2}} = \frac{1 - \epsilon_1}{\epsilon_1} + \frac{1}{X} + \frac{1 - \epsilon_2}{\epsilon_2}$$

对总热流率 Φ :

$$R_{\text{总,总}} = \frac{1 - \epsilon_1}{\epsilon_1 A} + \frac{1}{A X} + \frac{1 - \epsilon_2}{\epsilon_2 A}$$

注意: 这里是恰好均为 0.8, 要左到右算



空间辐射热阻网络单元

流程关键点复盘

步骤	核心动作	易错陷阱提醒
1. 找 L	确定特征长度	集总法用 V/A ; 单项近似解用几何尺度 (半厚度或半径)。
2. 判 Bi	选计算路径	别忘了 Bi 的 λ 是物体的导热系数, 不是流体的。
3. 查 μ, C	确定几何常数	确定你查的是圆柱表还是平壁表, 两者数值完全不同。
4. 算 θ	转换温度	公式里全是 θ 。最后交卷前检查: 有没有加回 t_f ?
5. 验 F_0	验证适用性	如果算出来的 F_0 特别小 (比如 0.01), 单项近似解无效。

$$\epsilon = \frac{\int_0^\infty \epsilon_\lambda E_{b\lambda} d\lambda}{E_b} = \epsilon_{\lambda,1} \cdot f(0 \rightarrow \lambda) + \epsilon_{\lambda,2} \cdot f(\lambda \rightarrow \infty)$$

Chp11 辐射: 黑体辐射分段计算

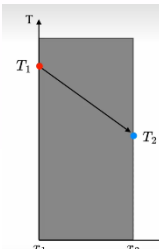
$$\epsilon \sigma A T^4 = -M c \frac{dT}{dT}$$

傅里叶定律

$$q = -k \frac{\Delta T}{\Delta x}$$

$$= -k \frac{(T_2 - T_1)}{x_2 - x_1}$$

$$= -k \frac{\Delta T}{\Delta x}$$

$$= k \frac{\Delta T}{\Delta x}$$


6 热辐射 & 黑体辐射公式

$$E_b = \sigma \times T^4$$

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4 \quad \text{黑体辐射常数}$$

$$q_b = \sigma(T_1^4 - T_2^4)$$

$$\varepsilon = \frac{E}{E_b} \quad \text{发射率 emissivity}$$

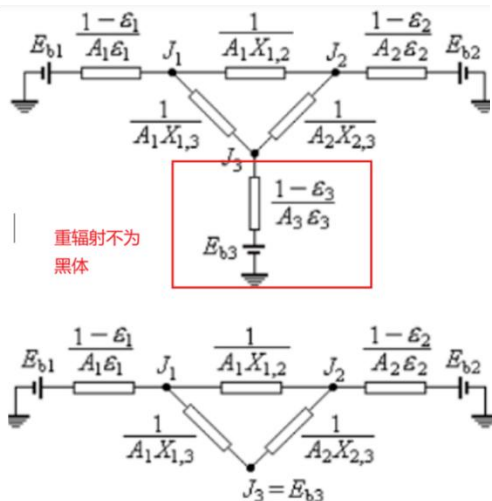
实际物体的发射率 epsilon

7 混合热阻的串/并联

$$q = q_1 + q_2 = \frac{T_1 - T_2}{\frac{1}{\sigma(T_1^2 + T_2^2)(T_1 + T_2)}} + \frac{T_1 - T_2}{\frac{1}{\varepsilon_1 \sigma}}$$

$$= \Delta T \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$= \frac{\Delta T}{\frac{R_1 R_2}{R_1 + R_2}}$$



根据基尔霍夫定律写出4个节点方程:

$$\frac{E_{b1} - J_1}{\frac{1 - \varepsilon_1}{\varepsilon_1 A_1}} + \frac{J_2 - J_1}{\frac{1}{A_1 X_{1,2}}} + \frac{J_3 - J_1}{\frac{1}{A_1 X_{1,3}}} = 0$$

■ 总换热量为对流换热量与辐射换热热量之和

$$\phi = \phi_c + \phi_r$$

$$= (h_c + h_r) A (t_w - t_f)$$

$$= h A (t_w - t_f)$$

例12-1 冬季一车间外墙内壁温度为 $t_w = 10^\circ\text{C}$; 内墙内壁温度为 $t_{\text{室}} = 16.7^\circ\text{C}$; 车间内气温 $t_f = 20^\circ\text{C}$ 。已知外墙内壁与内墙内壁的系统发射率为 $\varepsilon = 0.9$, 外墙内壁对流换热表面传热系数 $h_c = 3.21 \text{ W/(m}^2 \cdot \text{K)}$ 。求 (1) 外墙热流密度; (2) 外墙内壁复合换热表面传热系数; 热损失中辐射散热所占比例。

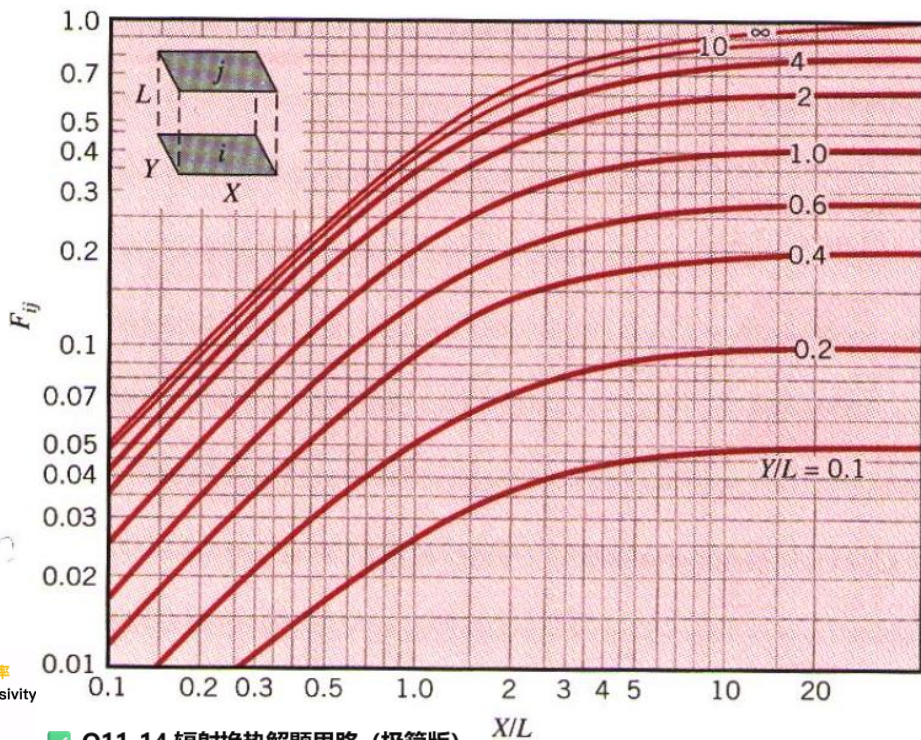
解:

$$h_r = \varepsilon \sigma_b (T_w^4 - T_f^4) / (t_f - t_w) = 3.21 \text{ W/(m}^2 \cdot \text{K)}$$

$$h = h_c + h_r = 6.42 \text{ W/(m}^2 \cdot \text{K)}$$

$$q = h(t_f - t_w) = 64.2 \text{ W/m}^2$$

$$h_r / h = 3.21 / 6.42 = 0.5$$



Q11-14 辐射换热解题思路 (极简版)

① 判题型

多表面 + 漫灰体 + 围壁 → 辐射热阻网络法

② 建模编号

1: 高温板 | 2: 低温板 | 3: 房间围壁 (黑体)

③ 基本量

- $A_1 = A_2 = 1$
- $\varepsilon_1 = \varepsilon_2 = 0.9, \varepsilon_3 = 1$
- $E_{bi} = \sigma T_i^4$

④ 角系数

- 平行正对: $F_{12} = F_{21}$ (查表)
- $F_{13} = 1 - F_{12}, F_{23} = 1 - F_{21}$

⑤ 辐射热阻

- 表面热阻: $R_{s,i} = \frac{1 - \varepsilon_i}{\varepsilon_i A_i}$
- 空间热阻: $R_{ij} = \frac{1}{A_i F_{ij}}$

igned parallel rectangles.

⑥ 列节点方程 (radiosity)

$$\frac{E_{bi} - J_i}{R_{s,i}} = \sum_j \frac{J_i - J_j}{R_{ij}} \quad ($$

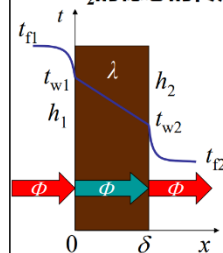
⑦ 净辐射量

$$Q_i = A_i \frac{E_{bi} - J_i}{R_{s,i}}$$

12-1 传热过程

一、通过平壁的传热

- 无内热源、导热率 λ 为常数、厚 δ 单层无限大平壁;
- 流体温度分别为 t_{f1} 与 t_{f2} 、表面传热系数分别为 h_1 与 h_2 的稳态的传热过程。



$$\phi = \frac{t_{f1} - t_{f2}}{\frac{1}{Ah_1} + \frac{\delta}{A\lambda} + \frac{1}{Ah_2}} = \frac{t_{f1} - t_{f2}}{R_{h1} + R_\lambda + R_{h2}} = \frac{t_{f1} - t_{f2}}{R_k}$$

$$\phi = Ak(t_{f1} - t_{f2}) = Ak\Delta t$$

$$\text{传热系数 } k = \frac{1}{\frac{1}{h_1} + \frac{\delta}{\lambda} + \frac{1}{h_2}}$$

$$\phi = \frac{t_{f1} - t_{f2}}{\frac{1}{\pi d_1 h_1} + \frac{1}{2\pi\lambda l} \ln \frac{d_2}{d_1} + \frac{1}{\pi d_2 h_2}}$$

$$\phi_i = \frac{\phi}{l} = \frac{t_{f1} - t_{f2}}{\frac{1}{\pi d_1 h_1} + \sum_{i=1}^n \frac{1}{2\pi\lambda_i} \ln \frac{d_{i+1}}{d_i} + \frac{1}{\pi d_{n+1} h_2}} \quad R_i \text{ 单位长度热阻}$$

上式写成

$$\phi = \pi d_2 l k_0 (t_{f1} - t_{f2}) = \pi d_2 l k_0 \Delta t$$

k_0 为以圆管外壁面面积为基准计算的传热系数, 计算式为:

$$k_0 = \frac{1}{\frac{d_2}{d_1} \frac{1}{h_1} + \frac{d_2}{2\lambda} \ln \frac{d_2}{d_1} + \frac{1}{h_2}}$$

问题: 如何确定 d_3 ?

$$R_i = \frac{1}{h_i \pi d_i} + \frac{1}{2\pi\lambda} \ln \frac{d_2}{d_i} + \frac{1}{2\pi\lambda_x} \ln \frac{d_3}{d_2} + \frac{1}{h_2 \pi d_3}$$

$$\frac{1}{h_2 \pi d_2} = \frac{1}{2\pi\lambda_x} \ln \frac{d_3}{d_2} + \frac{1}{h_2 \pi d_3}$$

$$d_3 = \exp \left(\ln d_2 + \frac{2\lambda_x}{h_2} (1/d_2 - 1/d_3) \right) \quad \text{迭代求解 } d_3$$

$$R_i = \frac{1}{h_i \pi d_i} + \frac{1}{2\pi\lambda} \ln \frac{d_2}{d_i} + \frac{1}{2\pi\lambda_x} \ln \frac{d_3}{d_2} + \frac{1}{h_2 \pi d_3}$$

$$\text{求极值: } \frac{dR_i}{dd_3} = \frac{1}{2\pi\lambda_x} \frac{1}{d_3} - \frac{1}{h_2 \pi d_3^2} = 0 \Rightarrow d_3 = d_c = \frac{2\lambda_x}{h_2}$$

总热阻达到极小值?

$$\frac{d^2 R_i}{dd_3^2} = \left(-\frac{1}{2\pi\lambda_x} \frac{1}{d_3^2} + \frac{2}{h_2 \pi d_3^3} \right) \bigg|_{d_3=d_c} = \frac{h_2^2}{8\pi\lambda_x^3} > 0 \Rightarrow \text{极小值}$$

